SMART INDIA HACKATHON 2026

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Problem Statement ID: **SIH26263**

Problem Statement Title: Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions

Theme: Smart Automation

PS Category: Software

Team ID: TCE_Yquanto

Team Name: **TCE_Yquanto (Thiagarajar College of Engineering, Madurai)**

Live Web Platform: **https://brainforge.info | https://pramaan.brainforge.info**

Live Backend & API: **https://apaar-id-based-system.onrender.com/docs**

Source Repository: **https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM**

Architecture Baseline: Next.js 14 + FastAPI + PostgreSQL (20 Tables, RLS) + Deterministic 3-Way Reconciliation

Verification Status: **66/66 Passing Tests | 20/20 E2E Checks | Live Cloud Deployment Verified**

DEPLOYED & VERIFIED SYSTEM ARCHITECTURE

PROPOSED SOLUTION: THE PRAMAAN-AICTE RECONCILIATION SYSTEM

1. REGULATORY CHALLENGE ADDRESSED

- **Fragmented Institutional Data:** Divergent college ERP spreadsheet schemas, non-standard column headers, and unverified declarations.
- **Reconciliation Bottlenecks:** Difficulties cross-verifying declared enrolments against actual student records and AICTE approval ceilings.
- **Intake & Duplicate Risks:** Lack of automated visibility into over-intake breaches, quota diversion, and concurrent multi-college claims.

2. WHAT PRAMAAN-AICTE IMPLEMENTS

- **Decoupled Institutional Portals:** Role-separated desks for College submission and Central AICTE regulatory governance.
- **Automated Ingestion Desk:** CSV/Excel file parsing with resilient column alias mapping into standard staging rows.
- **Deterministic 3-Way Reconciliation:** Pure Python comparison engine cross-checking declarations, staged student records, and approved capacity.
- **Persisted Validation Runs:** Stores row-level findings with precise classification: approved, quota breach, syntax error, or duplicate.

3. KEY WORKFLOW & GOVERNANCE VALUE

- **Explainable Row-Level Findings:** Direct, transparent feedback explaining specific discrepancy causes without opaque batch rejection.
- **Quarantine & Resubmission:** Interactive in-browser quarantine sandbox allowing institutions to correct data before council commit.
- **Controlled Central Decisions:** Council review workflow supporting formal decisions: approve, return for correction, or reject.
- **Audit Traceability:** Complete validation run history and append-only sequential SHA-256 audit chaining across actions.

AICTE CENTRAL APEX COMMAND & RECONCILIATION DASHBOARD

- **Central Intake Intelligence:** Live monitoring of sanctioned intake vs enrolled students across B.Tech, MCA, and Diploma.
- **Deterministic Review Desk:** Instant identification of seat limit breaches, unverified candidates, and duplicate claims.
- **Multi-Tenant Governance:** Seamless institutional isolation between AICTE Central Regulators, DTU Delhi, and COEP Pune.

Deterministic Python Core | 3-Way Cross-Reconciliation | Cryptographic Verification & Audit | 66/66 Verified Passing Tests

TECHNICAL APPROACH AND SYSTEM ARCHITECTURE

5-STAGE RECONCILIATION & GOVERNANCE PIPELINE

STAGE 1: Resilient File Ingestion

Upload institutional CSV/Excel files; dynamic column alias matching auto-normalizes divergent schemas into standardized staging records.

STAGE 2: Pure Python Validation Engine

Deterministic rule engine parses syntax, detects in-batch duplicates, verifies 12-digit APAAR format, and runs cross-field consistency checks.

STAGE 3: 3-Way Dynamic Reconciliation

Cross-compares declared student counts against verified stage records and sanctioned intake quotas from dynamic reference tables.

STAGE 4: Council Review & Audit Ledger

Central officers review row-level discrepancy findings; make formal Approve/Return decisions; actions chained into append-only SHA-256 audit ledger.

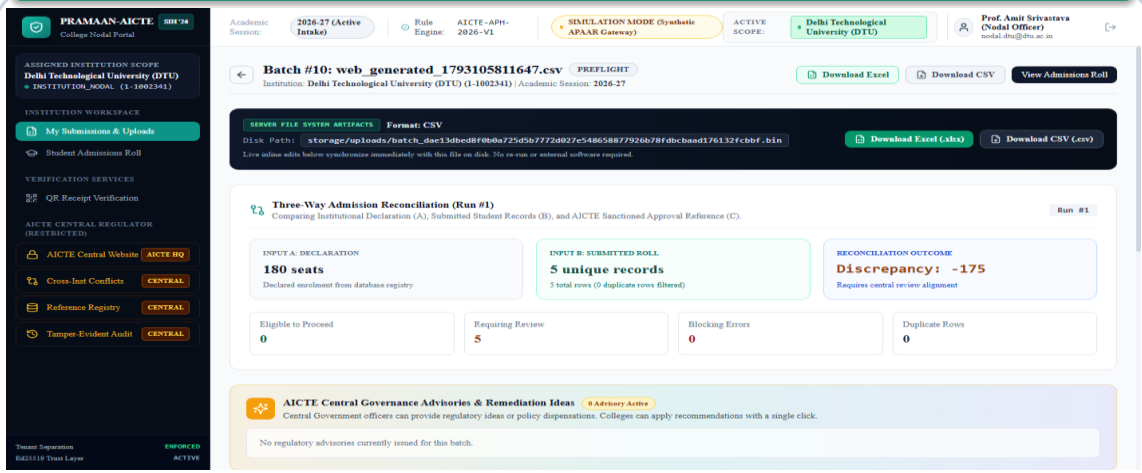
STAGE 5: Cryptographic Verification Receipt

Approved student records receive an asymmetric Ed25519 digital signature and portable verification receipt with offline verifiable QR code.

VERIFIED 4-LAYER SYSTEM ARCHITECTURE

- **Presentation Layer:** Next.js 14 (App Router), TypeScript, Tailwind CSS, Recharts for analytics; deployed on Vercel Edge.
- **Application & API:** FastAPI (Modular Monolith), Python 3.10+, Pydantic v2, Uvicorn, 73 endpoints; deployed on Render Cloud.
- **Persistence & Schemas:** SQLAlchemy ORM, Supabase PostgreSQL 17.11 (20 tables with RLS enabled) / SQLite WAL mode.
- **Cryptography & Security:** Ed25519 (RFC 8032 signing), Keyed HMAC-SHA256, sequential SHA-256 audit chaining.

INTERACTIVE QUARANTINE SANDBOX & ROW-LEVEL FINDINGS



Implemented UI: In-browser editable spreadsheet sandbox isolates validation findings before central commit.

FEASIBILITY, VIABILITY AND RISK MITIGATION

IMPLEMENTATION FEASIBILITY & VERIFIED MATURITY

- **Architectural Feasibility:** Zero-install web architecture. Standard modern browser access across Windows, Linux, and macOS. Decoupled Next.js client and FastAPI modular backend.

- **Verified Test Suite (66/66 Passing):** Exhaustive automated verification across 14 test suites covering authentication, institution onboarding, batch upload, deterministic reconciliation, security, and reference data.

- **Deterministic Logic Integrity:** Reconciliation logic is pure, explainable Python with explicit rule sets for sanctioned intake and supernumerary quotas; avoids opaque or non-deterministic AI decisions.

- **Synthetic Data Testing Engine:** Database-backed synthetic APAAR identities and editable approval/declaration models enable comprehensive pre-deployment stress testing and repeatable verification.

- **Dual-Dialect Database Support:** SQLAlchemy ORM architecture with dynamic driver fallback (psycopg3/psycopg2) verified on local SQLite WAL mode and Supabase PostgreSQL 17.11 (20 tables, RLS active).

- **Live Production Cloud Verification:** Full cloud deployment verified live: FastAPI backend on Render with PostgreSQL connection, Next.js frontend on Vercel with custom domain brainforge.info, and 100% end-to-end integration pass.

GOVERNANCE RISKS & TECHNICAL COUNTERMEASURES

- **Risk: Inconsistent Institutional Spreadsheets:** Colleges use varying ERP schemas and column headers (e.g. apaar_id vs uid vs student_apaar).
  * **Control:** Header alias normalizer maps 50+ divergent column variants into standard canonical schema automatically.

- **Risk: Student Identifier Privacy Exposure:** Storing plaintext 12-digit student APAAR identifiers risks data leakage under DPDP Act 2023.
  * **Control:** Keyed HMAC-SHA256 pseudonymized indexing and dynamic masking (XXXX-XXXX-9012) ensure zero plaintext PII exposure.

- **Risk: Multi-Institution Concurrent Admissions:** Students admitted concurrently across different institutions leading to seat blocking.
  * **Control:** Cross-institution collision radar cross-checks active admissions against centralized reference records and flags duplicates.

- **Risk: Regulatory Non-Repudiation & Tampering:** Unauthorized insider tampering with approved intake capacity or admission records.
  * **Control:** Append-only sequential SHA-256 hash chaining (Hash_n = SHA256(Record_n || Hash_n-1)) mathematically exposes any tampering.

- **Risk: Database Driver & Cloud Connection Mismatch:** Container deployment failures caused by PostgreSQL driver compatibility or pooler SSL configuration.
  * **Control:** Implemented smart connection URI sanitization and dynamic psycopg3/psycopg2 driver fallback with automated readiness health check.

MULTI-STAKEHOLDER IMPACT & DEGREE VERIFICATION

MULTI-STAKEHOLDER REGULATORY IMPACT

- **AICTE & Regulatory Councils:** Comprehensive real-time visibility into institutional enrolment versus sanctioned intake quotas; early detection of unauthorized excess enrolments and quota breaches.

- **Approved Institutions:** Standardized data submission, instant row-level discrepancy explanations, and controlled resubmission workflows replacing prolonged paper audit cycles.

- **Students & Deserving Candidates:** Transparent tracking protects legitimate seat allocations and safeguards affirmative action quotas from unauthorized commercial reallocation.

- **Academic Verifiers & Employers:** Fast offline verification via QR code and cryptographically signed verification receipts without requiring manual institutional registrar confirmation.

- **Social Equity & Affirmative Action:** Rigorous enforcement of statutory and supernumerary quotas (TFW, EWS, Kashmiri Migrants) against arbitrary substitution.

- **Administrative & Paperless Efficiency:** 100% digital data reconciliation replaces manual paperwork, physical affidavits, and physical institutional auditing logistics.

CRYPTOGRAPHICALLY VERIFIABLE DEGREE CERTIFICATE & QR PROOF

- **Dual Identity Display:** Certificates present Student Enrolment Roll No (ENR-108410E1) alongside masked APAAR ID (XXXXXXXX7890) and category quota status.

- **Asymmetric Ed25519 QR Code:** Embedded high-density QR code carries a digitally signed verification token; verifiers validate data integrity against AICTE public key offline.

- **Integrity Boundary Clarification:** Digital signature confirms platform-level record integrity and non-tampering; explicitly distinguished from official government identity confirmation.

RESEARCH, REFERENCES AND PROJECT RESOURCES

AUTHORITATIVE STANDARDS & REGULATORY CITATIONS

- AICTE Approval Process Handbook (APH 2026-27)

Statutory norms for approved institutional intake capacity, supernumerary seat limits (TFW, EWS, Kashmiri Migrants), and programme approval guidelines.

- National Education Policy (NEP 2020) & APAAR / ABC

Government policy framework establishing the 12-digit Automated Permanent Academic Account Registry as the lifelong national academic identifier.

- Digital Personal Data Protection Act (DPDP Act 2023)

Data fiduciary compliance, purpose limitation, identifier pseudonymization, dynamic masking, and student consent management frameworks.

- IETF RFC 8032: Ed25519 Cryptography Specification

Edwards-Curve Digital Signature Algorithm (Curve25519) providing mathematical non-repudiation and instant receipt verification.

- NIST SP 800-107: Recommendation for Secure Hashing

Cryptographic guidelines for sequential SHA-256 hash chaining utilized in the tamper-evident regulatory audit trail.

SCOPE DIFFERENTIATION (PRIOR ART VS PRAMAAN-AICTE)

- Validation Scope: Generic uploads check syntax only. PRAMAAN performs 3-way reconciliation (Declarations x Uploads x Sanctioned Capacity).

- Reference Data: Traditional workflows rely on static assumptions. PRAMAAN integrates dynamic database references with live revalidation.

- Finding Resolution: Standard systems use opaque batch rejections. PRAMAAN provides row-level explainable findings and an editable quarantine sandbox.

- Verification Output: Conventional portals yield unverified spreadsheets. PRAMAAN produces Ed25519-signed verification receipts and QR proofs.

VERIFIED PROJECT RESOURCES & LIVE DEPLOYMENT STATUS

- Live Web Platform: https://brainforge.info | https://pramaan.brainforge.info
(Production Vercel Edge with custom domain & Let's Encrypt SSL)

- Interactive Backend API & Docs: https://apaar-id-based-system.onrender.com/docs

(FastAPI OpenAPI 3.1 live documentation with 73 endpoints across 67 paths)

- Open-Source GitHub Repository: https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM

(Complete source code, 20 application models, Alembic migrations, test suites)

- Verified Test Suite: 66/66 pytest unit tests passing + 20/20 Phase F E2E checks + 10/10 live cloud integration tests verified.